

INDIAN STANDARD SPECIFICATION: IS 302 (PART 1): 2024

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES - GENERAL REQUIREMENTS

Standard Number: **IS 302 : Part 1 : 2024** (Harmonized with IEC 60335-1). Applicable to all single-phase household appliances rated up to 250 V.

MANDATORY SAFETY CLAUSES

- **Clause 8: Protection Against Access to Live Parts:** Test Probe B of IS 1401 applied with 10 N force must not contact hazardous live parts.
- **Clause 10: Power Input and Current:** Measured power input must not exceed marked rating by more than specified tolerances (+5% to +10%).
- **Clause 13: Electrical Strength & Leakage Current at Operating Temp:** Leakage current shall not exceed 0.75 mA for Class I portable appliances; high-voltage withstand of 1000 V to 1250 V AC.
- **Clause 19: Abnormal Operation & Fire Safety:** Appliances must not catch fire during stalled rotor or simulated electronic fault conditions. Glow Wire Test (IS 11000) at 750°C/850°C on insulating supports.

Important Note: Mandatory under the Electrical Appliances (Quality Control) Order. Applicable to water heaters (IS 302-2-201), electric irons (IS 302-2-3), and kitchen mixers (IS 302-2-14).